

Shakib Sadat Shanto

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EDUCATION

01/2020 – 01/2024

American International University-Bangladesh | Dhaka, Bangladesh | <https://www.aiub.edu/>

Bachelor of Science in Computer Science and Engineering

Grade (CGPA): 3.85 out of 4.0

PROFESSIONAL EXPERIENCE

08/2025 – Present

Researcher Engineer (NLP) | AIMS Lab, IRIIC, United International University (UIU) | Dhaka, Bangladesh | <https://iriic.uiu.ac.bd/>

- Conducting research on **AI for Healthcare**.
- Developing **LLM-driven** evaluation frameworks for **medical AI (Chatbots)** in low-resource languages (so far for Bengali).

02/2024 – Present

Lead Researcher (Remote) | Elite Lab | Cedarcroft Rd, Queens, NY, USA | <https://elitelab.ai/>

- Conducted research in **Natural Language Processing (NLP)**, focusing on **multilingual** and **multimodal** approaches.
- **Published** research works in reputable venues like **IEEE ACCESS** and **ICMLA**.
- Developed hybrid deep learning models with **Explainable AI** for emotion classification and hate speech detection.
- **Mentored** junior researchers and interns, providing guidance on research methodologies and best practices in AI development.

08/2023 – 01/2024

Research Assistant | American International University-Bangladesh | Dhaka, Bangladesh

- **Published three** journal research articles on **Natural Language Processing (NLP)**.
- **Published two** journal research articles on **Educational Technology**.
- Researched Generative AI's ability to promote **critical thinking** among undergraduate students using models like **Lee's thinking level** and **Bloom's Taxonomy**.
- Researched Generative AI's potential as a **teaching assistant** in **programming language courses**.

SELECTED PUBLICATIONS

Peer-reviewed full conference papers are prefixed with C, and peer-reviewed journal papers with J.

{C1} Morol, Md Kishor, **Shanto, Shakib Sadat**, Ahmed, Zishan, & Reza, Ahmed Shakib, (2024). **MultiLangMemeNet: A unified multimodal approach for cross-lingual meme sentiment analysis**. In Proceedings of 23rd International Conference on Machine Learning and Applications – ICMLA 2024, Miami, Florida. [Equal Contribution] [Code] [Link]

(C2) **Shanto, Shakib Sadat**, Paul, Rahul, Ahmed, Zishan, Reza, Ahmed Shakib, Islam, Kazi Mejbaul, & Roy, Saumya Shovan. (2024). **Console log explainer: A framework for generating automated explanations using LLM**. In Proceedings of 2nd International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings), Central Michigan University, USA. [\[Code\]](#) [\[Link\]](#)

(C3) **Shanto, Shakib Sadat**, Paul, Rahul, Reza, Ahmed Shakib, Islam, Kazi Mejbaul, & Roy, Saumya Shovan. **Bot vs. Bot: A Dual-Agent Approach for Testing Security in Conversational Systems**. ICMLA'25 [\[Under Review\]](#) [\[Code\]](#)

(C4) **Shanto, Shakib Sadat**, Morol, Md Kishor, Ahmed, Zishan, Reza, Ahmed Shakib, & Jubair, Md Abdullah Al. (2024). **Towards interpretable emotion classification in Bangla: A hybrid deep learning approach with explainable AI**. In Proceedings of 3rd International Conference on Computing Advancements - ICCA 2024. ACM. [\[Code\]](#) [\[Link\]](#)

(J1) Ahmed, Zishan, **Shanto, Shakib Sadat**, Rime, Most Humayra Khanom, Morol, Md Kishor, Fahad, Nafiz, Hossen, Md Jakir, & Jubair, Md Abdullah Al (2024). **The Generative AI Landscape in Education: Mapping the Terrain of Opportunities, Challenges and Student Perception**. IEEE Access (2024). [\[Link\]](#)

(J2) Reza, Ahmed Shakib, **Shanto, Shakib Sadat**, Zaima, Farah Ulfat, Sattar, Tazkera, Annisa, Mysha Maliha, Hossain, Muhammad Iqbal, Fahad, Nafiz, & Liew, Tze Hui. (2025). **Optimizing CNN Memory Efficiency: A Continual Learning Solution for Plant Stress Classification**. Nature Machine Intelligence. [\[Under Review\]](#) [\[Code\]](#)

(J3) **Shanto, Shakib Sadat.**, & Jony, A. I. (2025). **Interpretable Ensemble Learning Approach for Predicting Student Adaptability in Online Education Environments**. Knowledge, 5(2),10. [\[Link\]](#)

(C5) Ahmed, Zishan, Morol, Md Kishor, **Shanto, Shakib Sadat**, Reza, Ahmed Shakib, & Jubair, Md Abdullah Al. (2024). **Explainable English hate speech detection: A custom BiLSTM model with LIME for multilabel classification**. In Proceedings of 3rd International Conference on Computing Advancements - ICCA 2024. ACM. [\[Link\]](#)

(C5) Ahmed, Zishan, **Shanto, Shakib Sadat**, & Jony, Akinul Islam. (2024). **Unlocking Sentiment Analysis Potential for Low-Resource Bangla Language by Fine-Tuning LLaMA Model**. In Proceedings of the 27th International Conference on Computer and Information Technology (ICCIT) 2024. [\[Equal Contribution\]](#) [\[Link\]](#)

(C6) **Shanto, Shakib Sadat**, Ahmed, Zishan, & Jony, Akinul Islam. (2024). **Generative AI for programming education: Can ChatGPT facilitate the acquisition of fundamental programming skills for novices?** In Proceedings of 3rd International Conference on Computing Advancements - ICCA 2024. ACM. [\[Link\]](#)

(J6) **Shanto, Shakib Sadat**, Ahmed, Zishan, & Jony, Akinul Islam. (2023). **PAIGE: A generative AI-based framework for promoting assignment integrity in higher education**. STEM Education, 3(4), 288-305. [\[Link\]](#)

(J7) **Shanto, Shakib Sadat**, Ahmed, Zishan, & Jony, Akinul Islam. (2024). **Enriching learning process with generative AI: A proposed framework to cultivate critical thinking in higher education using Chat GPT**. Tuijin Jishu/Journal of Propulsion Technology, 45(1), 3019-3029. [\[Link\]](#)

(J8) Ahmed, Zishan, **Shanto, Shakib Sadat**, & Jony, Akinul Islam. (2024). **Potentiality of generative AI tools in higher education: Evaluating ChatGPT's viability as a teaching assistant for introductory programming courses**. STEM Education, 4(3), 165-182. [\[Link\]](#)

(C8) **Shanto, Shakib Sadat**, Reza, Ahmed Shakib, Islam, Safwan, Rafi, Rubiat, Sharmin, Momu, Ahmed, Zishan, & Jony, Akinul Islam. (2024). **From pixels to diagnosis: An innovative EfficientNetB3 approach for interpreting lung and colon cancer histopathology**. In Proceedings of the 27th International Conference on Computer and Information Technology (ICCIT) 2024. [\[Link\]](#)

SKILLS

- **Programming Languages:** Python, R, C#, Java, PHP, C++
- **Database System:** MySQL, Oracle
- **Frameworks:** Tensorflow, Pytorch, Langchain, RAG
- **Libraries:** Scikit-learn, Scikit-LLM, PEFT, LoRA NLTK, BNLTK, Transformers, OpenCV, Numpy, Pandas, Matplotlib
- **Tools & Software:** Github, Visual Studio, Weka, Tableau

PROJECTS

PDF Chatbot using LLM

- Developed a chatbot that can converse with PDFs, using **langchain** as memory retrieval and ADA-002 (OpenAI) model at its core.

Machine Learning Classification Web-App

- Developed a conventional ML classification **web app** and hosted on **Streamlit** where the performance of algorithms such as Naive Bayes, KNN, SVM, and Kernel SVM can be measured on different **hyperparameter** settings and can also be visualized using various **Scikit-learn** datasets.

Heart Failure Prediction in R

- Developed a heart failure prediction model using **KNN** in **R language** to predict heart diseases with various features of patients. The dataset was collected from Kaggle, and the model's performance was evaluated using an **accuracy** and **confusion matrix**.

Mining Student Data to Predict Result

- Conducted a performance analysis using **conventional ML** models to predict the academic performance of students using student data.

Game Store

- Followed **MVC pattern** while making webpages for game-sellers as part of a game store website using JS, PHP, CSS.

Language

- **IELTS**: Overall Band Score: 7.5; Listening: 8.5; Reading: 7.5; Writing: 6.5; Speaking: 6.5

AWARDS

Magna Cum Laude (Silver Medal)

- Awarded for maintaining a CGPA of **3.85** out of **4.0**.

The Prestigious Dean's Honor List

- Awarded dean's honor list in **4** semesters for achieving and maintaining a grade point average (CGPA) of **3.85** out of **4.0**.

REFERENCES

- **Dr. Akinul Islam Jony**, Postdoctoral Researcher, MeL and Game Research group, Universitat Oberta de Catalunya
Email: ajony@uoc.edu
- **Md. Kishor Morol**, Adjunct Professor, Department of CS, St. Thomas University
Email: mmorol@stu.edu
- **Saumya Shovan Roy**, Lead Engineer, Corelight, San Francisco
Email: saumya.roy@corelight.com